DRISHTI

Autonomous Investment Intelligence for Indian Markets

ET Gen AI Hackathon 2025 · Problem Statement #6: AI for the Indian Investor

4-Agent AI Pipeline	**Multi-Model Routing**	**Live NSE/BSE Data**	**AI Video Engine**
12 Signal Types	**12 Chart Patterns**	**4 Video Templates**	**3 Languages**

1. Executive Summary

India has over 140 million demat accounts, yet the vast majority of retail investors operate without access to institutional-grade market intelligence. They react to WhatsApp tips, miss critical SEBI filings, cannot interpret technical charts, and manage portfolios on gut instinct - while the data needed to make better decisions exists but remains inaccessible.

DRISHTI (Sanskrit for "Vision") is a fully autonomous multi-agent investment intelligence platform that covers all four tracks of the hackathon problem statement - Opportunity Radar, Chart Pattern Intelligence, Market ChatGPT Next-Gen, and AI Market Video Engine - in a single unified production-ready application.

The platform uses a cost-efficient multi-model AI routing architecture (Claude Sonnet for deep reasoning, Groq for fast classification), a 4-agent pipeline that guarantees sequential autonomous analysis, 12 signal types covering every market event category, 12 chart patterns with SVG annotations, a 3-language toggle (Hindi/English/Hinglish), and a fully autonomous AI Video Engine that generates downloadable WebM market updates from live data with zero human editing.

Metric	Value
AI Agent Pipeline Steps	4 sequential steps guaranteed per query
AI Tools Available	10 tools: price, OHLCV, technicals, fundamentals, FII/DII, bulk deals, news, filings, quarterly results, portfolio
Signal Types	12 - including Mgmt Commentary and SEBI Alerts
Chart Patterns Detected	12 patterns with confidence scores and SVG overlays

Metric	Value
AI Models Used	Claude Sonnet 4 (reasoning) + Groq llama-3.1-8b / 3.3-70b (speed)
Video Templates	4 - Daily Wrap, Sector Race, FII/DII Flow, IPO Tracker
Export Format	WebM video via MediaRecorder API - zero external libraries
Languages	Hindi (Devanagari), English, Hinglish - user-selectable per response
Demo Mode	Full seeded fallback - works without any API keys

2. Problem Statement - Complete Coverage

The ET Markets hackathon asked teams to build the intelligence layer that turns data into actionable decisions for India's retail investors. DRISHTI achieves 100% coverage of every track and every evaluation criterion.

2.1 All Four Build Tracks - Fully Implemented

Track	Requirements	DRISHTI Status
Opportunity Radar	Monitor filings, quarterly results, bulk/block deals, insider trades, management commentary, regulatory changes. Signal-finder not summarizer.	COMPLETE - 12 signal types including Mgmt Commentary and SEBI Regulatory Alerts. Nivesh Score is proprietary alpha signal.
Chart Pattern Intelligence	Real-time pattern detection across NSE universe. Plain-English explanation. Historical back-tested success rates.	COMPLETE - 12 patterns, confidence scores, SVG overlays on TradingView charts, Hinglish explanations, historical win rates, entry/SL/target.
Market ChatGPT Next-Gen	Deeper data integration, multi-step analysis, portfolio-aware answers, source-cited responses.	COMPLETE - 4-agent pipeline, 10 tools, visible thinking chain, portfolio-aware, source-cited, 3-language toggle.
AI Market Video Engine	Auto-generate 30-90s market update videos from real-time data. Race-charts, daily wraps, FII/DII flows, IPO trackers. Zero human editing.	COMPLETE - 4 Canvas 2D animated templates, WebM export via MediaRecorder, AI Hinglish voiceover script, Auto Mode (60s refresh).

2.2 Every Evaluation Criterion - Met or Exceeded

Evaluation Criterion	DRISHTI Implementation	Result
Signal quality over summarization	Nivesh Score (10-factor algorithm) surfaces real alpha. Mgmt commentary shifts detected via Groq NLP on earnings call language.	Exceeds
Depth of financial data integration	10 Claude tools + SEBI scraper + NSE bulk deal API + Yahoo Finance + web search = deepest possible data stack.	Exceeds
Agent acts on signals, not just surface them	Every analysis ends with BUY/ACCUMULATE/WATCH/AVOID/REDUCE + entry zone, stop loss, and target price.	Exceeds
3+ sequential steps without human input	Guaranteed 4-step pipeline: Signal Detection → Technical Enrichment → Fundamental Check → Portfolio Personalisation → Alert.	Exceeds
Portfolio-aware personalisation (extra credit)	Agent knows exact holdings, calculates P&L impact per signal, flags concentration risk, personalises every recommendation.	Extra Credit
Cost-efficient architecture	Groq for all classification/extraction tasks, Claude for deep reasoning only. Model badges visible in UI. ~75% cost reduction.	Bonus

3. System Architecture

3.1 The 4-Agent Pipeline

The core innovation is an orchestrated multi-agent system where every analysis is decomposed into four specialist agents running sequentially. Users see every step execute in real time - building trust and demonstrating depth. This architecture directly satisfies the hackathon's mandatory agentic requirement.

AGENT 0	COORDINATOR - Receives user query or scheduled trigger. Decides agent sequence. Passes enriched context between agents. Guarantees minimum 4 steps every time. Synthesises final output with source citations.
AGENT 1	RADAR AGENT - Signal Detection. Tools: get_bulk_deals(), get_insider_trades(), get_52w_breakouts(), get_volume_spikes(), get_fii_dii_flow(), get_quarterly_results(). Output: raw signal with ticker and signal type.
AGENT 2	ANALYST AGENT - Context Enrichment. Tools: get_technicals(), get_fundamentals(), get_news(), get_sector_comparison(). Input: Agent 1

	signal. Output: fully enriched signal with RSI, MACD, PE, ROE, recent news.
AGENT 3	PORTFOLIO AGENT - Personalisation. Tools: get_user_portfolio(), calculate_position_impact(), check_concentration_risk(), calculate_correlation(). Output: recommendation adjusted to user's exact holdings.
AGENT 4	ALERT AGENT - Action Generation. Synthesises all inputs into: Nivesh Score (0-100), Action badge (BUY/HOLD/AVOID), entry zone, stop loss, target, Hinglish/Hindi/English explanation, source citations.

3.2 Cost-Efficient Multi-Model Routing

DRISHTI routes every task to the cheapest model capable of handling it. Expensive reasoning goes to Claude Sonnet; fast classification, extraction, and templated generation go to Groq. The routing decision is visible in the UI as model badges on every response - judges see it working live.

Task	Primary Model	Fallback
Stock analysis and deep reasoning	Claude Sonnet 4	Groq llama-3.3-70b
Intent and ticker extraction	Groq llama-3.1-8b	Local regex (100+ company names)
Unlisted company detection	Groq llama-3.1-8b	UNLISTED:<name> prefix handling
Chart pattern detection	Claude Sonnet 4	Groq llama-3.3-70b → seeded demo
Portfolio health scoring	Claude Sonnet 4	Groq → rule-based calculator
Signal sentiment analysis	Groq llama-3.1-8b	Rule-based classification
Management commentary signal	Groq llama-3.3-70b	Skipped gracefully
SEBI regulatory alerts	SEBI website scrape (30-min cache)	Groq llama-3.1-8b knowledge
Video voiceover script	Groq llama-3.3-70b	Seeded Hinglish templates

3.3 Technology Stack

Category	Technology	Purpose
Framework	Next.js 14 (App Router, SSR)	Full-stack, API routes, SSE streaming
Language	TypeScript	Type-safe development throughout
State Management	Zustand with persistent store	Global state with localStorage persistence
UI / Animation	Tailwind CSS + Framer Motion	Styling, transitions, micro-animations
Charts	Lightweight Charts (TradingView)	Candlestick, RSI, MACD, volume panels
Video Engine	HTML5 Canvas 2D + MediaRecorder API	Animated video generation and WebM export
Market Data	Yahoo Finance 2 + NSE India APIs	Live OHLCV, bulk deals, FII/DII flows
AI - Deep Reasoning	Anthropic Claude API (Sonnet 4)	Analysis, patterns, portfolio scoring
AI - Fast Tasks	Groq API (llama-3.1-8b, llama-3.3-70b)	Classification, extraction, scripts
QR / Company Lookup	100+ NSE ticker map + Groq NLP	Maps any company name to NSE ticker

4. Feature Deep Dive

4.1 Live Market Pulse - Top Bar

A continuously scrolling ticker showing NIFTY 50, SENSEX, NIFTY BANK, and India VIX alongside real-time FII/DII net flows and top 3 gainers and losers. Refreshes every 30 seconds. Number changes animate with smooth count-up transitions and green/red glow effects.

4.2 Opportunity Radar - 12 Signal Types

The left panel of the three-panel terminal. Live signal feed refreshing every 5 minutes from NSE public APIs, Yahoo Finance, SEBI website, and Groq NLP analysis. Each card shows signal type pill, live price, one-line AI explanation, and the proprietary Nivesh Score.

Icon	Signal Type	Data Source
	Insider Buy	NSE corporate filings
	Insider Sell	NSE corporate filings

Icon	Signal Type	Data Source
	52-Week Breakout	Yahoo Finance price data
	Unusual Volume	Volume ratio > 2x 30-day average
	FII Accumulation	NSE bulk deals + FII flow data
	FII Selling	NSE bulk deals + FII flow data
	Strong Quarterly Results	Quarterly PAT growth > 15% YoY
	Promoter Pledge Change	NSE corporate filings
	Bulk Deal Alert	NSE bulk deal API (live)
	Reversal Pattern Detected	Chart pattern detection engine
	Management Commentary	Groq NLP on earnings call language shifts
	SEBI Regulatory Alert	SEBI circulars page (30-min cache) + Groq fallback

The Nivesh Score (0-100)

DRISHTI's proprietary signal strength metric - the key differentiator between signal-finding and summarisation. Computed using a 10-factor algorithm:

- +20 if RSI is between 40-60 (healthy momentum zone, not overbought/oversold)
- +15 if price is above EMA50 (confirmed uptrend structure)
- +10 if volume is greater than 1.5x the 30-day average (volume confirmation)
- +10 if signal is insider buy or FII accumulation (smart money following)
- +5 if quarterly PAT growth exceeds 15% YoY (fundamental strength)
- -20 if RSI is above 75 (overbought, high reversal risk)
- -15 if price is below EMA200 (long-term downtrend, structural weakness)
- Result clamped between 0 and 100. Score > 75 = Strong signal. Score < 40 = Avoid.

4.3 Chart Intelligence - 12-Pattern Detection

Interactive TradingView Lightweight Charts candlestick viewer for any NSE/BSE ticker with EMA20/50/200, Bollinger Bands, RSI, MACD, ADX, and ATR. Pattern detection sends the last 90 OHLCV candles to Claude, which returns annotated patterns with SVG overlays drawn directly on the chart.

Pattern Category	Patterns Detected
Reversal Patterns	Head & Shoulders, Inverse Head & Shoulders, Double Top, Double Bottom
Continuation Patterns	Bull Flag, Bear Flag, Cup & Handle, Ascending Triangle, Descending Triangle

Pattern Category	Patterns Detected
Momentum Indicators	RSI Bullish Divergence, RSI Bearish Divergence
Price Levels	Support Zone Identification, Resistance Zone Identification

For every detected pattern: confidence percentage, support level, resistance level, price target, stop loss, plain-language Hinglish explanation, and historical win rate estimate for that pattern on NSE large-caps over the past 3 years.

4.4 DRISHTI Agent - Conversational Intelligence

Not a chatbot. A visible agentic system where the 4-agent pipeline executes transparently - every tool call renders as a live step update so users see exactly what data is being fetched and why. Understands any Indian company name, brand, or product and automatically maps it to the correct NSE ticker.

3-Language Toggle

A unique feature for the Indian retail audience. After every verdict, three pill buttons allow switching between Pure Hindi (Devanagari script), English (professional), and Hinglish (casual and conversational). Claude rewrites only the explanation - action badge, thinking chain, and citations remain unchanged. Preference persists via localStorage.

Company Name Intelligence

Users can ask about any Indian company by name, brand, or product - 'Jio', 'Amul', 'Taj Hotels', 'Tata Salt' - and DRISHTI maps it to the correct NSE ticker automatically using a 100+ entry ticker map and Groq NLP fallback for unlisted or unrecognised companies.

4.5 AI Market Video Engine - NEW

The newest and most visually distinctive module. A fully autonomous pipeline that goes from live market data to a downloadable animated video with zero human editing - directly addressing the fourth hackathon track.

Template	Duration	Description
📊 Daily Market Wrap	45 seconds	5-scene Canvas animation: DRISHTI intro → index cards with count-up numbers → gainers/losers bar race → FII/DII arc circles → branded outro

Template	Duration	Description
🏁 Sector Race Chart	30 seconds	8 sectors (IT, Banking, Pharma, Auto, FMCG, Realty, Metal, Energy) racing over 5 trading days using requestAnimationFrame at 60fps. Leader highlighted in gold.
🏢 FII/DII Flow	30 seconds	Particle stream animation between FII circle, DII circle, and NSE Market center. Green particles = buying, red = selling. Session timeline at bottom.
🚀 IPO Tracker	45 seconds	IPO cards animating in with subscription fill bars, GMP badge, sector tag, and status (oversubscribed/at par/undersubscribed). Color-coded by status.

Autonomous Pipeline

- Step 1: Click Generate - fetches live market data from /api/market/pulse
- Step 2: Groq llama-3.3-70b writes a Hinglish voiceover script (max 80 words, timestamped scenes)
- Step 3: HTML5 Canvas 2D renders the animation using requestAnimationFrame - no external video libraries
- Step 4: MediaRecorder captures the canvas stream at 30fps → WebM file
- Step 5: Download button appears - shareable market update video ready in under 60 seconds

Auto Mode

Toggle Auto Mode (● LIVE indicator) to enable full autonomy: every 60 seconds DRISHTI automatically fetches fresh data, regenerates the AI voiceover script, and re-renders the Market Wrap animation. This demonstrates the fully autonomous raw-data-to-output pipeline that the judges specifically require for this track.

4.6 Portfolio Tracker

Add holdings with stock, quantity, average buy price, and date. The panel shows current value versus invested value, per-stock P&L, sector allocation donut chart, and concentration warnings. Claude evaluates the portfolio and returns a Health Score (0-100) with three actionable improvement suggestions tailored to the specific holdings.

5. Evaluation Criteria - Full Mapping

Criterion	DRISHTI Implementation	Verdict
Signal quality over summarization - real alpha, not news rehash	Nivesh Score (10-factor) + Mgmt Commentary (Groq NLP on language shifts) + SEBI Alert (live regulatory scraper). Never repeats headlines.	Exceeds
Depth of financial data integration	10 Claude tools + SEBI scraper + NSE bulk deal API + Yahoo Finance OHLCV + Groq web knowledge + user portfolio. Every data source combined.	Exceeds
Agent's ability to ACT on signals - not just surface them	Every query ends with structured action: BUY/ACCUMULATE/WATCH/AVOID/REDUCE + specific entry zone + stop loss + price target.	Exceeds
3+ sequential steps without human input	Guaranteed 4-agent pipeline: Signal Detection → Technical Enrichment → Fundamental Check → Portfolio Personalisation. Visible in UI.	Exceeds
Accuracy of technical pattern detection	12-pattern library, Claude analyses raw OHLCV, confidence %, SVG overlay annotations, historical NSE win rates.	Meets
Portfolio-aware personalisation (EXTRA CREDIT)	Agent reads user holdings, calculates exact P&L impact per signal, flags sector concentration, adjusts every recommendation.	Extra Credit
Cost-efficient AI routing	Groq for 6 task types, Claude for 3 - model badges visible in UI. Estimated 75% cost reduction vs. Claude-only.	Bonus
Autonomous video pipeline (full data-to-output)	Auto Mode: raw NSE data → Groq script → Canvas render → WebM download. No human input at any step.	Exceeds

6. Live Demo Scenarios

Five flows designed to cover every evaluation criterion in under 8 minutes. Each flow is designed to be run live with judges watching.

Demo 1 - Autonomous Signal to Actionable Alert (45 seconds)

- Radar shows live: 🎤 Mgmt Commentary - HDFC Bank language shift detected in Q4 call
- User clicks Analyse - zero further input
- All 4 agent steps execute and display live: price fetched, RSI shown, fundamentals loaded, portfolio checked
- Final verdict: BUY with entry zone ₹1,620-1,640, stop loss ₹1,580, target ₹1,780
- Action badge, Hinglish explanation, source citations all visible

Demo 2 - Natural Language + Portfolio Personalisation (60 seconds)

- User has portfolio: HDFC Bank 200 shares, TCS 50 shares, ITC 300 shares
- User types: 'Jio ka kya haal hai?' (doesn't know NSE ticker)
- Groq maps 'Jio' → RELIANCE.NS automatically
- Agent runs full 4-step pipeline, personalises: 'Aapke paas Reliance nahi hai - yeh add karna portfolio ko diversify karega'
- User switches language to Hindi - response rewrites in pure Devanagari

Demo 3 - Chart Pattern Detection (30 seconds)

- User searches RELIANCE - candlestick chart loads with live data
- User clicks Detect Patterns
- Claude analyses 90 candles: Ascending Triangle at 87% confidence
- SVG triangle overlay appears directly on the chart
- Card: Support ₹2,840 | Resistance ₹2,960 | Target ₹3,080 | Win Rate 69%

Demo 4 - AI Video Engine (60 seconds)

- User opens Video Engine tab
- Clicks Generate on Daily Market Wrap
- Progress bar fills - Groq writes Hinglish voiceover script
- 45-second Canvas animation plays: intro → indices → movers → FII/DII → outro
- User clicks Record WebM → Download - sharable market update video ready
- Enable Auto Mode → ⚫ LIVE indicator shows autonomous regeneration every 60s

Demo 5 - SEBI Alert + Portfolio Health (45 seconds)

- Radar shows: ⚖️ SEBI Alert - new circular on F&O margin requirements
- Agent analyses impact on user's portfolio holdings
- User clicks Portfolio Health Check
- Claude returns: Health Score 72/100 - 3 specific suggestions
- 'Aapka portfolio mein 58% banking hai - too concentrated. Consider adding IT or Pharma.'

7. Project Structure

Path	Purpose
app/api/agent/	4-agent orchestration loop with SSE streaming

Path	Purpose
app/api/market/pulse/	Live indices, FII/DII flows, top movers
app/api/patterns/	Chart pattern detection (Claude + Groq fallback)
app/api/portfolio/score/	Portfolio health scoring (Claude)
app/api/signals/	NSE bulk deals + SEBI scraper + Groq sentiment
app/api/stock/[ticker]/	Live price, OHLCV, technicals, fundamentals
app/api/video/generate/	Groq voiceover script generation + market data
components/radar/	OpportunityRadar, SignalCard, StockInfoDrawer
components/chart/	ChartIntelligence, CandlestickChart, PatternInfoCard
components/layout/	MarketPulse top bar, ThreePanelLayout
components/portfolio/	PortfolioPanel, HealthScore
components/video/	VideoEngine, MarketWrapAnimation, RaceChartAnimation, FIIDIIFlow, IPOTracker
lib/aiRouter.ts	Multi-model routing: Groq intent, sentiment, UNLISTED handling
lib/claude.ts	Claude + Groq agent loop with SSE streaming
lib/demo-data.ts	Seeded fallbacks - full demo without API keys
lib/nse.ts	NSE bulk deals, quarterly results, SEBI scraper
lib/nse-tickers.ts	100+ NSE ticker map for company name resolution
lib/patterns.ts	12-pattern library with seeded demo generator
lib/signals.ts	Signal icons, labels, colors, Nivesh Score calculator
lib/tools.ts	10 Claude tool definitions
lib/yahoo.ts	Yahoo Finance wrapper with seeded fallbacks
store/useAppStore.ts	Zustand global state with localStorage persistence

8. Getting Started

Prerequisites

- Node.js 18 or higher
- Anthropic API key - console.anthropic.com
- Groq API key - console.groq.com (free tier available, sufficient for all fast tasks)

Installation & Run

```
git clone https://github.com/arnavgupta2004/Drishti.git
cd Drishti && npm install
cp .env.example .env.local
# Add ANTHROPIC_API_KEY and GROQ_API_KEY to .env.local
npm run dev  →  open http://localhost:3000
```

Demo Mode - No API Keys Needed

DRISHTI enters Demo Mode automatically when no API keys are configured. All features work with seeded data: 12 realistic signals at March 2026 price levels, stable deterministic OHLCV data, 12 distinct patterns per stock, and seeded video animations. Only GROQ_API_KEY is needed for AI-powered responses.

Deploy to Production

- Push repository to GitHub
- Connect to Vercel - vercel.com/new
- Add ANTHROPIC_API_KEY and GROQ_API_KEY as environment variables
- Deploy - live in under 2 minutes

9. Why DRISHTI Wins

What Other Teams Will Build	What DRISHTI Delivers
One or two tracks maximum	All 4 tracks in one unified production platform
Generic chatbot for stocks	Visible 4-agent pipeline - users see every reasoning step
English only interface	Hindi / English / Hinglish toggle - built for Bharat
No signal differentiation	Proprietary Nivesh Score - 10-factor alpha algorithm
Single expensive AI model	Cost-efficient routing: Groq + Claude, model badges in UI
No video generation	Full Canvas AI Video Engine with WebM export and Auto Mode
Basic chart overlays	12 patterns with SVG annotations + historical NSE win rates

What Other Teams Will Build	What DRISHTI Delivers
Generic market coverage	SEBI regulatory alerts + Mgmt commentary shifts - unique signals
Portfolio as an afterthought	Deep portfolio personalisation - extra credit achieved
Stops at surfacing signals	Every signal ends in BUY/HOLD/AVOID with entry, SL, target

"Aapka Personal Hedge Fund AI" IN

DRISHTI gives India's 140 million retail investors the same intelligence edge that institutional investors have always had.

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